

COS 226

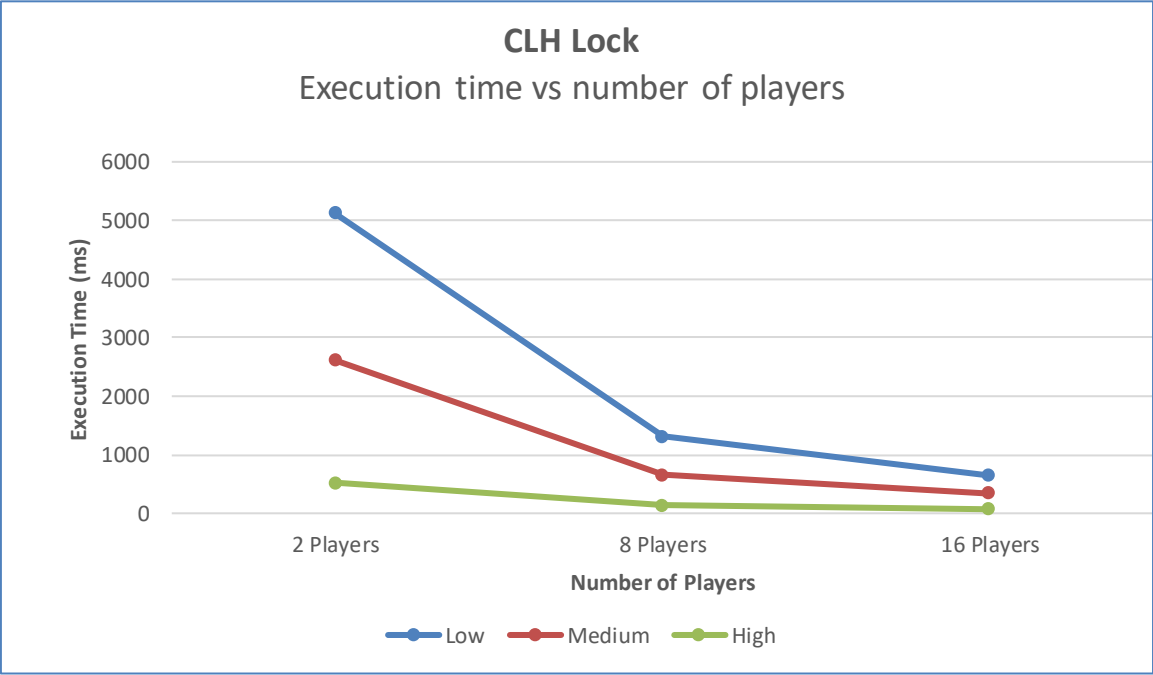
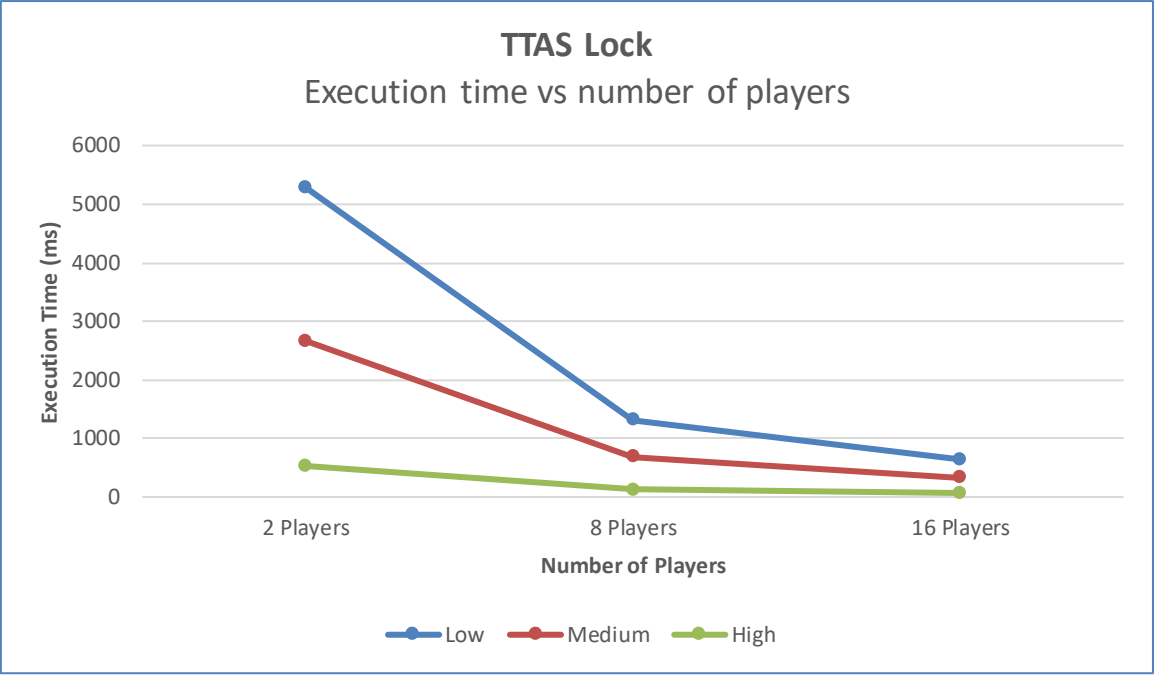
Assignment 1 Report

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How the low, medium and high contention was simulated

To simulate the different levels of contention I changed the thinkMs and work values for each level of contention as follows:

- **Low – (thinkMs = 10, work = 1000)**
This gave the players a longer think time and a short work time so that fewer players tried to access the chest at the same time and they released the lock quickly.
- **Medium – (thinkMs = 5, work = 5000)**
Players had a slightly shorter think time and a longer work time which created more overlap since players would arrive more frequently to wait for the chest, and they held the lock longer.
- **High – (thinkMs = 1, work = 5000)**
Players had a very short think time here, which caused constant competition for the lock since many players arrived almost simultaneously and stayed in the critical section for longer.



Differences in execution time vs number of players for TTAS and CLH lock

For both the TTAS and CLH lock, the execution time would consistently decrease as the number of players increases. With 2 players, execution times are the highest at around 5 seconds, with 8 players the execution times drop to around 1 second and with 16 players they fall below 1 second.

The reason why I think this happens is because with more players the coins are taken in parallel by multiple threads, so all the coins are taken from the chest much faster. With only 2 players each player would need to take nearly 1000 coins which takes longer. With 16 players the workload is divided a lot more evenly between players and the game finishes much quicker.

Overall, when comparing the two locks, they both show very similar trends in execution time but there are some small differences. Since TTAS uses backoff it reduces wasted spinning while CLH queues the thread in a fair order. Looking at my results, CLH was slightly faster under high contention, because its queue lets each thread wait on its own spot instead of all spinning on the same lock.

Time difference for each lock in different contention

In low contention both the TTAS and CLH perform similarly with 2, 8 and 16 players, however the TTAS lock is slightly slower (5281ms vs 5115ms). This is because TTAS threads still spin directly on the shared lock variable creating extra traffic, while CLH threads spin on their own local nodes, reducing interference.

In medium contention the execution time decreases significantly as the players increase and both locks perform very close to each other with TTAS being slightly faster (693ms vs 654ms). The overlap of players trying to access the chest more often increases competition, but neither is heavily impacted.

In high contention the difference between the locks becomes clearer. TTAS execution times drop quickly with more players, but CLH is slightly faster at 16 players. This is because CLH will queue threads in order and prevent them from spinning on the same lock.

Which lock is the fairest?

The CLH lock is the fairest since it uses a queue structure. Each thread needs to wait its turn in a FIFO order so no player can take the lock many times in a row. All players get mostly an equal chance to take coins from the chest.

The TTAS lock is less fair because the threads can keep trying to get the lock after a short backoff which means some threads may get the lock more often than others which means the coins are distributed a bit more unevenly than the CLH lock

Which lock scales the best as the players increase?

Both locks get faster as the number of players increases because more threads can take coins at the same time. TTAS scales slightly better under low and medium contention since backoff reduces the wasted spinning. But overall, the CLH lock scales the best under high contention because its queue prevents threads from constantly spinning on the same variable, keeping the system fairer. Therefore, CLH is the best choice as the number of players increases.